

Generative Imputation with Multi-level Causal Consistency for Variable Subset Forecasting

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Abstract

Variable Subset Forecasting (VSF) poses critical challenges in time series analysis when entire variables become unavailable during inference. Existing imputation methods relying on inter-variable correlations fail catastrophically in VSF due to two inherent limitations: (1) *Missing variable collapse*, where the complete absence of certain variables invalidates correlation-based dependency learning, and (2) *Temporal covariate shift*, where time-evolving data distributions destabilize correlation patterns learned from training data. To address these fundamental issues, we propose Generative Imputation with Multi-level Causal Consistency (GIMCC), establishing causality-driven imputation as the first principled solution for VSF. Our key innovation lies in enforcing **causal invariance** through dual consistency constraints: global causal isomorphism ensures the imputed variables preserve the ground-truth causal graph structure of the complete system, while local causal subgraph alignment maintains consistency between observed variables and their causal neighborhood dependencies. By decoupling causality from spurious correlations, GIMCC provides time-invariant imputation signals robust to distribution shifts, which explicitly preserves causal relationships via multivariate spectral convolutions. Extensive experiments across five real-world domains demonstrate that GIMCC achieves average improvements of 20–60% in MAE/RMSE over correlation-based imputation baselines, remarkably outperforming full-variable training (*Oracle*) in temporal covariate shift scenarios. Our work bridges the critical gap between causal analysis and practical forecasting systems under variable absence, offering theoretically grounded guarantees for real-world deployment.

CCS Concepts

• Information systems → Spatial-temporal systems.

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KDD '25, Toronto, ON, Canada

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ACM ISBN 979-8-4007-1454-2/2025/08
<https://doi.org/10.1145/3711896.3736980>

Keywords

Variable Subset Forecasting, Granger Causality

ACM Reference Format:

Qi Hao, Yue Gao, Runchang Liang, Yunhe Zhang, and Pengyang Wang. 2025. Generative Imputation with Multi-level Causal Consistency for Variable Subset Forecasting. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '25)*, August 3–7, 2025, Toronto, ON, Canada. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3711896.3736980>

KDD Availability Link:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.15504834>.

1 Introduction

Variable Subset Forecasting (VSF) is a distinctive and challenging problem in time series forecasting when only a subset of the variables is available during the inference phase, while others are missing. In typical missing data scenarios, only a few scattered data points are absent, allowing models to infer based on the majority of the available data. In contrast, VSF involves the complete absence of specific variables during inference, compelling the model to operate with a significantly reduced subset of the original variable set. For example, consider a network of weather stations that monitor various environmental factors such as temperature, humidity, and wind speed. During extreme weather conditions, some stations might fail due to equipment damage, resulting in a total loss of data from those locations. This absence of critical data can significantly impact the forecasting accuracy for the entire monitoring network, as the model must make predictions without the complete set of variables it was trained on. The uniqueness of VSF lies in the necessity for models to maintain high accuracy despite substantial gaps in the input data. The challenge extends beyond handling sporadic missing data points to performing reliable forecasting with entirely missing variables. This requires advanced imputation techniques and robust forecasting models capable of leveraging the available subset of data to make accurate predictions.

Existing data imputation methods mainly rely on the correlations or similarities between variables to estimate missing values [1, 3, 19, 28, 33], which operates under the assumption that missing values

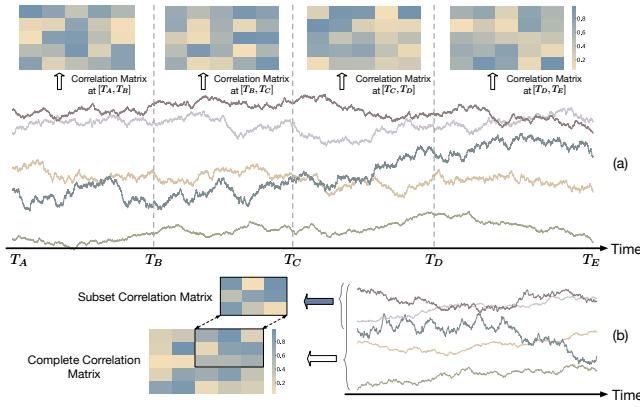


Figure 1: An example of the correlation between time series variables, where the heat map matrix represents the correlation matrix. The darker the color, the tighter the correlation. (a) demonstrates that correlation will vary over time as a result of covariate shift. (b) indicates that the existing correlation will also change because of unseen variables.

can be inferred from observed patterns in the data by leveraging statistical dependencies or learned representations. For example, SAITS [3] utilizes two weighted self-attention blocks to capture temporal dependencies and feature correlations for data imputation; FDW [1] recovers the missing variables during the inference phase by selecting top-K nearest neighbors in the training dataset.

However, inter-variable correlations and similarities are unreliable for imputation in the VSF setting due to several critical reasons. **(i) Missing variable collapse:** In VSF, the complete absence of certain variables results in the lack of both intra-variable correlation and inter-variable dependency. This absence hinders effective imputation of missing variables and negatively impacts prediction performance [15]. **(ii) Covariate shift over time:** Covariate shift is prevalent in multivariate time series [23], meaning that the correlations between variables differ between the training data (complete observations) and the testing data (variable subset). Consequently, applying correlations from the training data to the test data for imputation is problematic, as this mismatch significantly degrades imputation performance [11].

Given these challenges, we propose utilizing causality as the guiding principle for imputation in VSF instead of traditional correlations or similarities. Causality offers a stable and invariant signal, remaining consistent over time and under different conditions. This stability makes causality a reliable foundation for imputing missing variables, ensuring that the imputed data aligns more accurately with the true underlying relationships. Nevertheless, integrating causality in VSF requires addressing two unique characteristics: **(i) Global imputed-graph causality consistency**, which ensures that the causal graph derived from the imputed data is isomorphic to the causal graph obtained from the complete data. This means that the overall structure of causal relationships is preserved, providing a comprehensive and accurate representation of dependencies among variables. **(ii) Local subset-subgraph causality consistency**, which focuses on ensuring that the causal relationships

within the available variable subset are maintained as a subgraph of the complete data’s causal graph. This local consistency is crucial for accurately reflecting the causal influences within the limited data subset available during inference. By incorporating these two levels of causal consistency, we ensure that the imputed data not only fits well with the observed data but also adheres to the true underlying causal mechanisms, leading to more accurate and reliable forecasting in the VSF setting.

Along this line, we introduce Generative Imputation with Multi-level Causal Consistency (GIMCC), a novel framework designed to address the challenges of VSF by transferring multivariate time series into a linear time-invariant system for imputation, coupled with global and local causal consistency constraints. GIMCC takes the available variable subset as input and generates imputed data by mapping the data to the frequency space and learning a comprehensive embedding space for the variables. To integrate multi-level causalities, we formulate the global and local causality consistency guarantees as graph and subgraph isomorphism tasks. This approach regulates the learned embeddings of variables to ensure compliance with the causal graph isomorphism between the complete variable set and the imputed variables, as well as the subgraph isomorphism between the complete variables and the available variable subset. By maintaining these causal consistencies, GIMCC ensures that the imputed data aligns with the true underlying causal relationships. To further enhance VSF performance, we adopt a joint learning strategy that simultaneously trains the imputation and forecasting models. This concurrent training allows the models to enhance each other’s performance, leading to more accurate and reliable forecasting outcomes.

In summary, our proposed GIMCC provides a robust solution to the unique challenges of VSF by leveraging the stability of causal relationships and enforcing multi-level causal consistency, significantly improving imputation and forecasting accuracy. Our key contributions can be listed as follows:

- **Integration of Causality into VSF:** By incorporating causal relationships, we address the limitations of correlation-based imputation methods, enhancing the accuracy and reliability of the imputed data.
- **Novel Causality-induced Imputation Method:** We map multivariate time series into a time-invariant system and reconstructs inter-variable causal relations in the frequency domain, leveraging the stability of causality for accurate imputation.
- **Global and Local Causality Consistency:** We introduce and enforce causal consistency constraints at both global imputed-graph and local subset-subgraph levels, ensuring that the imputed data adheres to true causal mechanisms.
- **Comprehensive Evaluations:** Extensive evaluations on multiple datasets demonstrate the effectiveness of our framework in addressing the unique challenges of VSF, showcasing its superiority over existing methods.

2 Preliminaries

2.1 Causal Discovery

Given the N -variate observations $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$, where $\mathbf{x}_i$ denotes the i -th variable, the causal relationship between these N variables can be represented by a Directed Acyclic Graph (DAG) $\mathcal{G} = (V, E)$,

where V denotes the variable (node) set and E denotes the edge set for causal relations between variables. Specifically, if variable x_i causes the variable x_j , we assume that variable i is one of the parent nodes of variable j and there is a directed edge between x_i and x_j : $(x_i \rightarrow x_j) \in E$. The goal of causal discovery is to learn a mapping function

$$\mathcal{F} : \mathbf{X} \rightarrow \mathcal{G} \quad (1)$$

that can derive the DAG $\mathcal{G}$ from the observed data $\mathbf{X}$.

2.2 Variable Subset Forecasting

Given an L length past time series data $\{\mathbf{x}_{(i)}^{t-L:t}\}_{i=1}^N = [\mathbf{x}_{(i)}^{t-L}, \dots, \mathbf{x}_{(i)}^t]$, multi-variate time series forecasting aims to infer the subsequent Q -length data $\{\mathbf{x}_{(i)}^{t:t+Q}\}_{i=1}^N = [\mathbf{x}_{(i)}^{t+1}, \dots, \mathbf{x}_{(i)}^{t+Q}]$.

In the context of time series forecasting, we introduce the variable subset forecasting problem. Let Ψ_N represent the complete set of N -variate observations, and let Ψ_S denote an S -variate subset, where $\Psi_S \subset \Psi_N$ and $S \ll N$. In this problem setting, during the inference phase, only the subset Ψ_S is available, whereas the complete data Ψ_N is accessible during the training phase. The objective of variable subset forecasting is to develop a forecasting model trained on the complete data Ψ_N that can generalize effectively to the subset Ψ_S .

3 Methodology

In this section, we introduce our framework in detail. We start with the overall framework. Then, we present the details of each component in GIMCC.

3.1 Framework Overview

Figure 2 shows an overview of our proposed framework GIMCC. Overall, GIMCC imputes missing variables regulated by reconstructing inherent inter-variable causal relations from different perspectives. GIMCC employs a causality-induced variable imputation method, which takes a subset of variables as input and outputs a complete set of imputed variables. During the process, the imputation method is forced to reconstruct the inter-variable causality on two levels: (i) **Global DAG Consistency**, that the DAG of imputed variables should be isomorphic to the DAG of the real complete variables; (ii) **Local DAG Consistency**, that the DAG of the given variable subset should be subgraph-isomorphic to the DAG of the real complete variables. The imputed variables are then fed into downstream forecasting backbones. GIMCC optimizes the imputation and forecasting models jointly, ensuring coherent integration and improved performance.

3.2 Causality-induced Variable Imputation

For multivariate time series, the data generation process is fundamentally governed by causal relationships between variables. As these causal relationships are inherently stable, the interactions between different variables remain consistent over time. If the multivariate time series is conceptualized as a system, the state of the system depends solely on changes in the underlying causal relationship. This stability allows us to model multivariate time series as a linear time-invariant system which possesses time-invariant characteristics. In an LTI system, the relationship between inputs and

outputs can be represented through convolution, where the output is derived by convolving the input with a function that characterizes the system's properties. Within the framework of multivariate time series, the causal graph serves as this characteristic function. Consequently, by convolving the input data (variable subset) with the causal graph, we can impute the missing variables. The convolution theorem [22] states that the convolution of two signals in the time domain corresponds to the product of their Fourier transforms in the frequency domain. Therefore, the convolution of a variable subset with the complete causal graph can be transformed into the multiplication of the two vectors in the frequency domain.

We detail the design as follows. For simplicity, we omit the time span superscript for the variables. Formally, let $\mathbf{X}_S = \{\mathbf{x}_i | \forall \mathbf{x}_i \in \Psi_S\}$ denote the available subset. In the training phase, given an L -length time period, the subset Ψ_S is selected by randomly masking a certain percentage of the original variables Ψ_N , where $\mathbf{X}_S \in \mathbb{R}^{S \times L \times D}$ and D denotes the dimension of the variable.

Specifically, we set the values of the missing variables to zero at their corresponding positions and get the new input data as $\{\mathbf{U} = \mathbf{X} | \{\mathbf{X}_{N-S}\} = 0\}$. And then, we impute the missing variables in the frequency domain through Fast Fourier Transform (FFT) based on the causal graph $\mathcal{G}$:

$$\begin{aligned} \hat{\mathbf{X}} &= \mathbf{U} * \mathcal{G} = \mathcal{F}^{-1}(\mathcal{F}(\mathbf{U}) \times \mathcal{F}(\mathcal{G})), \\ \text{FFN}(\hat{\mathbf{X}}) &= \text{Dropout}(\tilde{W}_2(\text{ReLU}(\tilde{W}_1 \hat{\mathbf{X}}) + \tilde{b}_1) + \tilde{b}_2), \\ \hat{\mathbf{X}}_N &= \text{LayerNorm}(\hat{\mathbf{X}} + \text{Dropout}(\text{FFN}(\hat{\mathbf{X}}))), \end{aligned} \quad (2)$$

where $*$ denotes the convolution operator. $\tilde{W}$ and $\tilde{b}$ represent the weight and bias terms respectively. $Y_k = \mathcal{F}(y_i) = \sum_{i=0}^T y_i \times e^{-j \frac{2\pi}{T} ki}$ and $y_i = \mathcal{F}^{-1}(Y_k) = \frac{1}{T} \sum_{k=0}^{T-1} Y_k \times e^{j \frac{2\pi}{T} ki}$ represent Fourier transform and inverse Fourier transform respectively, where T represents the length of input data.

Therefore, the loss function of the imputation backbone $\mathcal{L}_{\text{IMP}}$ can be represented as:

$$\mathcal{L}_{\text{IMP}} = \frac{1}{N} \|\mathbf{X} - \hat{\mathbf{X}}_N\|, \quad (3)$$

where $\mathbf{X}$ and $\hat{\mathbf{X}}_N$ represent the real complete data and the imputed data, respectively.

3.3 Global and Local Causality Consistency Guarantee

In this section, we will explain how to obtain the causal graph $\mathcal{G}$ and ensure both global and local causality consistency.

3.3.1 Preliminary Causal Discovery. Granger causality is a popular tool for analyzing the causal characteristics of time-series data in many real-world applications. We adopt cMLP [30] as the base model for preliminary causal discovery. As Granger causality analysis [7] is a powerful method that determines cause and effect based on predictability, which reflects the influence of one variable j on the predictive relationship with another variable i . Therefore, the causal discovery task can be formulated as a prediction task. cMLP utilized a multilayer perceptron (MLP) [32] to model multivariate time series data [30]. However, due to the black-box nature of neural networks, it is challenging to determine the specific weights. For example, we can not specify the weight that allows time series j to

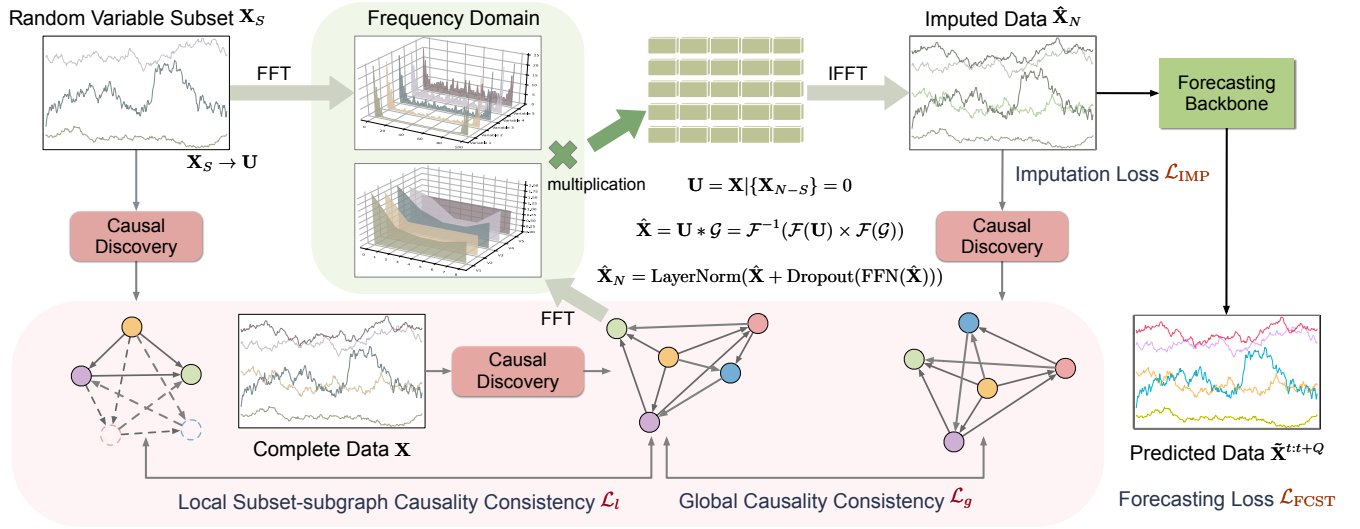


Figure 2: Framework Overview. **Top: The main architecture.** For N -variant time series $X = \{x_i | \forall x_i \in \Psi_N\}$, certain variables are randomly masked to obtain a variable subset $X_S = \{x_i | \forall x_i \in \Psi_S\}$, where $\Psi_S \subset \Psi_N$ and $S \ll N$. The subset Ψ_S is then fed to the causality-conditioned generative imputation model to obtain a reconstructed time series $\hat{X}_N$. Subsequently, the reconstructed time series is used for the forecasting task, ultimately yielding the prediction results $\hat{X}^{t:t+Q}$. **Bottom: Global and Local Causality Consistency Guarantee (pink).** For the global causality consistency, the DAG of $\hat{X}_N$ should be approximately isomorphic to the DAG of X . For the local subset-subgraph causality consistency, the DAG of X_S should be a subgraph of the DAG of X .

Granger cause time series i but not Granger causes time series q for $q \neq i$. Therefore, we model each variable with a separate model [30] instead of modelling g jointly for all variables $\mathbf{x}^t = \{x_i^t\}_{i=1}^N$:

$$x_i^t = g_i(\mathbf{x}^{<t_1}, \dots, \mathbf{x}^{<t_p}) + e_i^t, \quad (4)$$

where $g_i(\cdot)$ represents the relationship that how the other variables affect the variable i and e_i^t represents additive zero mean noise.

Let $g_i(\cdot)$ denote an MLP with L layers and h_l^t represent the values of the d -dimensional l -th hidden layer at time step t . The weights and biases of the neural networks at each layer are defined as $\mathbf{W} = \{W_1, \dots, W_L\}$ and $\mathbf{b} = \{b_1, \dots, b_L\}$, respectively. For different time lag $m \in [1, M]$ at the first layer, the weight can be represented as $W_1 = \{W_{11}, \dots, W_{1M}\}$. Therefore, the hidden values of the first layer can be represented as

$$h_1^t = \sigma\left(\sum_{m=1}^M W_{1m} \mathbf{x}^{t-m} + b_1\right), \quad (5)$$

where σ is the activation function. And for each layer of the neural network, the hidden units can be given by

$$h_l^t = \sigma(W_l h_{l-1}^t + b_l). \quad (6)$$

So, we can get the final output

$$x_i^t = g_i(\mathbf{x}^{<t}) + e_i^t = W_L h_L^t + b_L + e_i^t. \quad (7)$$

For the first layer weight matrix W_{1m}^j , if the j -th column contains zeros for all m , time series j does not Granger cause time series i . The sparse weight matrix represents the causal relationship between different variables, which corresponds to the final causal graph $\mathcal{G}$. By aggregating all time points, we derive the final average causal graph $\mathcal{G} = \text{CD}(\mathbf{X})$ for the input data $\mathbf{X} = \sum_{t=1}^T \mathbf{x}^t$.

3.3.2 Global Imputed-graph Causality Consistency Guarantee. In addition to pointwise reconstruction, maintaining inter-variable causal consistency is a crucial indicator of imputation quality. Specifically, we formulate the causality consistency between the real data X and imputed $\hat{X}_N$ as a graph isomorphism task, where we regulate the DAG of $\hat{X}_N$ to be approximately isomorphic to the DAG of X , referring to global causality consistency guarantee.

To this end, we first derive the ground-truth DAG $\mathcal{G}$ of the real complete data X . Then, to further achieve the isomorphism, we leverage the Weisfeiler-Lehman (WL) kernel [27] to constrain the similarity between $\mathcal{G}$ and $\hat{\mathcal{G}}$. The WL kernel computes the similarity between graphs by iteratively refining node labels (initialized as node degree) based on the structure of their neighborhoods and then comparing these refined labels. It measures how similar two graphs are by comparing the distributions of these refined labels, providing a quantitative measure of graph similarity. Specifically, for a node v in $\mathcal{G}$, its label $\Gamma^i(v)$ after the i -th iteration is updated by

$$\Gamma^i(v) = \text{HASH}(\Gamma^{i-1}(v), \{\{\Gamma^{i-1}(u) : u \in \text{Nbr}(v)\}\}), \quad (8)$$

where HASH is a hash function used to generate a unique new label, $\text{Nbr}(v)$ denotes the set of neighboring nodes of node v , and $\{\{\cdot\}\}$ denotes a multiset, which includes repeated elements.

After completing all iterations, the WL kernel similarity between two graphs $\mathcal{G}$ and $\hat{\mathcal{G}}$ is given by

$$K(\mathcal{G}, \hat{\mathcal{G}}) = \sum_{i=0}^I \sum_{v \in \mathcal{G}} \sum_{w \in \hat{\mathcal{G}}} \delta(\Gamma^i(v), \Gamma^i(w)), \quad (9)$$

where δ is the Kronecker delta function and I is the total number of iterations. The objective of global causality consistency is to ensure

the isomorphism between the DAGs $\mathcal{G}$ and $\hat{\mathcal{G}}$, i.e., maximizing the kernel similarity $K(\mathcal{G}, \hat{\mathcal{G}})$, which is equivariant to minimizing the loss function:

$$\mathcal{L}_g = e^{-K(\mathcal{G}, \hat{\mathcal{G}})}. \quad (10)$$

3.3.3 Local Subset-Subgraph Causality Consistency Guarantee.

Definition 3.1. Subgraph Isomorphism. Given two graphs $\mathcal{G}_A = (V_A, E_A)$ and $\mathcal{G}_B = (V_B, E_B)$, where V s and E s are node and edge set, respectively. $\mathcal{G}_A$ is isomorphic to one subgraph of $\mathcal{G}_B$, if and only if (1) $V_A \subseteq V_B$, and (2) $E_A \subseteq E_B$ and all the endpoints in E_A are in V_A .

PROPOSITION 1. *In the variable subset forecasting setting, the causal DAG of a given variable subset is isomorphic to one subgraph of the DAG of the complete variables.*

PROOF. Let the causal DAG for the complete set of variables be denoted as $\mathcal{G} = (\Psi_N, E_N)$ and the causal DAG for the variable subset as $\mathcal{G}_S = (\Psi_S, E_S)$, where E_N and E_S represent the directed edge sets (causal relationships) of $\mathcal{G}$ and $\mathcal{G}_S$, respectively. In VSF, the available variable set Ψ_S is a subset of the complete variable set Ψ_N , i.e., $\Psi_S \subseteq \Psi_N$. Missing variables do not imply the absence of causal influence; rather, the observations for these variables are inaccessible due to operational constraints. Thus, the causal influences of the missing variables on the observed ones remain intact, and the missing observations do not alter the actual causal relationships among the remaining variables. Consequently, the causal relationships between the variables in the subset Ψ_S are preserved as in the original causal DAG. Formally, $E_S \subseteq E_N$ and all endpoints of the edges in E_S are contained within Ψ_S . This satisfies the definition of subgraph isomorphism. Therefore, $\mathcal{G}_S$ is isomorphic to a subgraph of $\mathcal{G}$. $\square$

We first employ an l layer encoder to capture the relationships both within and between variables, generating a stable node embedding. Each layer contains a feed-forward network, with residual connections and layer normalization applied between layers.

$$\text{FFN}(\mathbf{X}) = \hat{W}_2(\hat{W}_1\mathbf{X} + \hat{b}_1) + \hat{b}_2, \quad (11)$$

$$\text{Enc-Layer}(\mathbf{X}) = \text{LayerNorm}(\mathbf{X} + \text{FFN}(\mathbf{X})), \quad (12)$$

where $\hat{W}$ and $\hat{b}$ represent the weight and bias terms respectively. The embeddings for variable subset and the complete data can be denoted as

$$\mathbf{h}_S = \text{Enc}(\mathbf{X}_S) = \sum_{i=1}^l \text{Enc-Layer}_i(\mathbf{X}_S), \quad (13)$$

$$\mathbf{h} = \text{Enc}(\mathbf{X}) = \sum_{i=1}^l \text{Enc-Layer}_i(\mathbf{X}). \quad (14)$$

And then, we employ a Graph Neural Network (GNN) to transform the subset variable embeddings $\mathbf{h}_S$ into the DAG embedding

$$\mathbf{z}_S = \text{GraphPooling}(\text{GNN}(A_S, \mathbf{h}_S)), \quad (15)$$

where A_S and $\mathbf{h}_S$ represent the adjacency matrix and the node features $\mathcal{G}_S$. We derive the DAG embedding $\mathbf{z}$ of the complete data following the same operations:

$$\mathbf{z} = \text{GraphPooling}(\text{GNN}(A, \mathbf{h})), \quad (16)$$

where A and $\mathbf{h}$ represent the adjacency matrix and the node features of $\mathcal{G}$, respectively.

Then, inspired by NeuroMatch [35], we adopt the order embedding space to constrain the DAG embeddings of $\mathbf{z}_S$ and $\mathbf{z}$ to follow subgraph isomorphism: if $\mathcal{G}_S$ is a subgraph of $\mathcal{G}$, then the embedding $\mathbf{z}_S$ of the DAG $\mathcal{G}_S$ should be in the “lower-left” of the embedding $\mathbf{z}$ of the DAG $\mathcal{G}$ in the embedding space:

$$\mathbf{z}_S[r] \leq \mathbf{z}[r] \forall r=1^R \quad \text{iff} \quad \mathcal{G}_S \subseteq \mathcal{G}, \quad (17)$$

where R is the embedding dimension. The order embedding space can be preserved by optimizing the following max margin loss

$$E(\mathbf{z}_S, \mathbf{z}) = \|\max\{0, \mathbf{z}_S - \mathbf{z}\}\|_2^2, \quad (18)$$

$$\mathcal{L}_l = \sum_{(\mathbf{z}_S, \mathbf{z}) \in \text{Pos}} E(\mathbf{z}_S, \mathbf{z}) + \sum_{(\mathbf{z}_S, \mathbf{z}) \in \text{Neg}} \max\{0, \theta - E(\mathbf{z}_S, \mathbf{z})\}, \quad (19)$$

where Pos denotes the set of positive examples (real subset), and Neg denotes the set of negative samples. We construct negative samples by perturbing the directed edges in the DAG. We set θ (a hyperparameter) as a threshold to ensure a minimum separation between positive and negative samples.

3.4 Model Training & Inference

3.4.1 Training Phase. The goal of variable subset forecasting is to accurately predict the available subset. To this end, we use the forecasting model to regularize the imputation model. Unlike traditional approaches, we adopt a joint learning strategy, where both the imputation and forecasting models are trained simultaneously. During this phase, both models are updated concurrently, enabling mutual influence and resulting in more accurate predictions.

Let $\tilde{\mathbf{X}}^{t:t+Q}$ denote the predicted results generated by the forecasting model, which takes the imputed data $\hat{\mathbf{X}}_N$ as input. We use Mean Absolute Error (MAE) as the forecasting loss:

$$\mathcal{L}_{\text{FCST}} = \frac{1}{S} \|\mathbf{X}^{t:t+Q} - \tilde{\mathbf{X}}^{t:t+Q}\|. \quad (20)$$

Therefore, the overall loss function of the joint-learning can be represented as

$$\mathcal{L} = \mathcal{L}_{\text{FCST}} + \alpha \cdot \mathcal{L}_{\text{IMP}} + \beta \cdot \mathcal{L}_g + \gamma \cdot \mathcal{L}_l, \quad (21)$$

where α , β and γ are hyperparameters to control the contributions of generative imputation backbone and global & local causality, respectively. We optimize the proposed framework by minimizing the joint learning loss $\mathcal{L}$ via gradient descent.

3.4.2 Inference Phase. During the inference phase, we apply the well-trained model GIMCC to accommodate the new test variable subset for forecasting. Initially, this subset is fed into the generative imputation backbone, which subsequently generates the complete set of variables. These generated variables are then utilized as input for the forecasting model to carry out the inference. Consequently, we extract the predicted results corresponding to the available subset, thus presenting them as the final output for the VSF task.

4 Experiment

In this section, we conducted extensive experiments to evaluate the effectiveness of the proposed model. And our experiments aim to answer the following research questions (RQ):

Table 1: An illustration of the overall performance of time series forecasting on subset variables. The results are reported as an average and standard deviations (values in parenthesis) of 10 runs. Each run corresponds to one trained model with 100 epochs, and each epoch involves a randomly sampled variable subset. Δ_{subset} represents the improvement of our model over the *Partial* setting. The larger the value, the better the performance for the Δ_{subset} .

Forecasting Model		METR-LA		SOLAR		ECG5000		ETTh1		ETTm1	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
MTGNN	Partial	4.54(0.37)	8.90(0.68)	4.36(0.53)	6.04(0.81)	3.88(0.61)	6.54(1.10)	2.02(1.17)	3.34(2.01)	1.30(0.76)	2.17(1.32)
	Oracle	3.49(0.25)	7.21(0.50)	2.94(0.27)	4.66(0.57)	3.43(0.54)	5.94(1.08)	1.65(0.90)	2.79(1.61)	1.01(0.52)	1.72(0.93)
	Ours	3.25(0.23)	6.55(0.43)	2.38(0.22)	3.58(0.41)	3.14(0.51)	5.42(1.01)	1.57(0.89)	2.64(1.64)	0.96(0.55)	1.65(1.03)
	Δ_{subset}	28.41%	26.40%	45.41%	40.73%	19.07%	17.13%	22.28%	20.96%	26.15%	23.96%
ASTGCN	Partial	5.57(0.72)	10.61(1.36)	6.14(1.29)	8.95(2.35)	3.60(0.60)	6.05(1.13)	2.21(1.10)	3.44(1.98)	1.37(0.65)	2.21(1.18)
	Oracle	5.04(0.39)	9.59(0.62)	4.54(0.47)	6.48(0.85)	3.47(0.50)	5.83(0.99)	1.98(1.09)	3.27(1.93)	1.01(0.52)	1.72(0.93)
	Ours	3.74(0.25)	7.31(0.43)	2.57(0.23)	3.86(0.45)	3.29(0.47)	5.58(0.98)	1.61(0.93)	2.69(1.71)	0.98(0.57)	1.68(1.06)
	Δ_{subset}	32.85%	31.10%	58.14%	56.87%	8.61%	7.77%	27.15%	21.80%	28.47%	23.98%
MSTGCN	Partial	4.78(0.43)	9.35(0.75)	4.75(0.73)	7.02(1.42)	4.34(0.87)	7.61(1.86)	2.22(1.08)	3.44(1.93)	1.34(0.62)	2.24(1.11)
	Oracle	4.49(0.31)	8.93(0.50)	3.64(0.41)	5.60(0.82)	3.39(0.52)	5.82(1.06)	1.91(1.05)	3.17(1.86)	1.15(0.61)	1.97(1.09)
	Ours	3.72(0.26)	7.35(0.45)	2.38(0.22)	3.69(0.43)	3.12(0.51)	5.50(1.05)	1.60(0.94)	2.66(1.70)	0.97(0.56)	1.67(1.05)
	Δ_{subset}	22.18%	21.39%	49.89%	47.44%	29.57%	27.73%	27.93%	22.67%	27.61%	25.45%
TGCN	Partial	9.92(0.75)	15.66(0.94)	8.76(0.87)	12.15(1.63)	6.22(1.37)	9.91(2.27)	3.75(1.65)	5.54(2.62)	1.71(0.79)	2.60(1.37)
	Oracle	8.57(0.92)	14.78(1.27)	4.56(0.78)	7.32(1.64)	6.16(1.29)	9.84(2.20)	3.69(1.55)	5.44(2.44)	1.52(0.74)	2.44(1.28)
	Ours	5.43(0.40)	10.03(0.58)	3.48(0.36)	5.02(0.60)	5.10(1.20)	8.28(2.26)	1.91(0.96)	3.06(1.84)	1.11(0.59)	1.82(1.12)
	Δ_{subset}	45.26%	35.95%	60.27%	58.68%	18.01%	16.45%	49.07%	44.77%	35.09%	30.00%
Average Improvement		32.18%	28.71%	53.43%	50.93%	18.82%	17.27%	31.61%	27.55%	29.33%	25.85%

- **RQ1:** Does the proposed model exhibit efficacy in addressing the Variable Subset Forecasting (VSF) task?
- **RQ2:** Can our proposed model achieve performance beyond the upper bound of VSF when evaluated on a complete variable set with no missing data?
- **RQ3:** How does GIMCC compare to existing imputation techniques in terms of VSF predictive superiority?
- **RQ4:** Do the causality consistency terms truly benefit the final forecasting performance on variable subset?
- **RQ5:** How do the weights of multiple loss functions affect the overall performance of the model? Is there an optimal combination of these weights?
- **RQ6:** Does joint learning truly benefit the final performance?

4.1 Experimental Setup

4.1.1 Data Description. We conduct our experiments on five real-world datasets: *METR-LA* [14], *SOLAR*¹, *ECG5000*², *ETTh1* [40] and *ETTm1* [40]. The detailed contents of these datasets will be introduced in Appendix A.1.

4.1.2 Backbone Algorithms and Imputation Baselines. After the missing variable imputation, we employ various forecasting models as the backbone for the subsequent forecasting task: MTGNN [34], ASTGCN [8], MSTGCN [8] and TGCN [39], where detailed information will be introduced in Appendix A.2.

To fully evaluate the performance of the proposed method, we compare our framework with the below data imputation baselines, including traditional imputation methods (KNNE [2], IIM [38], TRMF [37], CSDI [31], SAITS [3], SS-GAN [19] and MICE [33]), and latest VSF methods (TRF [12], TOI-VSF [10] and FDW [1]).

4.1.3 Comparison Setting. For each of the forecasting backbones described above, we implement two distinct settings. While both settings share an identical training process that utilizes the entire dataset, they differ in their respective inference stages.

- **Partial.** We completely remove the data for the variables $N - S$ during the test stage and do not perform any data imputation for these missing variables. This setting illustrates the forecasting model's original performance on VSF.
- **Oracle.** The *Oracle* setting represents an idealized case of the VSF task only for comparison. In this setting, data in Ψ_N is assumed to be all available during the inference phase. And, the loss metrics are computed exclusively for the variables in Ψ_S to align with our comparison objectives.

4.1.4 Evaluation Metrics. We evaluate performance using two commonly used metrics in multivariate time series forecasting tasks: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Furthermore, we introduce two new metrics to compare the effectiveness of our method with the two settings described above.

- **Partial Improvement (Δ_{subset}).** Δ_{subset} is used to quantify the performance of the proposed method for the *Partial* setting,

¹<https://www.nrel.gov/grid/solar-power-data.html>

²<http://www.timeseriesclassification.com/description.php?Dataset=ECG5000>

Table 2: Relative performance to *Oracle* setting. $\Delta_{improve}$ indicates the proportion by which our model surpasses the *Oracle* setting. For the $\Delta_{improve}$, the larger the value, the better the performance.

Forecast Model		METR-LA		SOLAR		ECG5000		ETTh1		ETTm2	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
ASTGCN	$\Delta_{improve}$	25.79%	23.77%	43.39%	40.43%	5.19%	4.29%	18.69%	17.74%	19.01%	18.84%
MSTGCN	$\Delta_{improve}$	17.15%	17.69%	34.62%	34.11%	7.96%	5.50%	16.23%	16.09%	15.65%	15.23%
MTGNN	$\Delta_{improve}$	6.88%	9.15%	19.05%	23.18%	8.61%	7.77%	4.85%	5.38%	4.95%	4.07%
TGCN	$\Delta_{improve}$	36.64%	32.14%	23.68%	31.42%	17.21%	15.85%	48.24%	43.75%	26.97%	25.41%
Average Improvement		21.61%	20.09%	30.18%	32.28%	9.70%	8.60%	22.99%	20.74%	16.65%	15.89%

defined as the follows:

$$\Delta_{subset} = \frac{E_{partial} - E_{ours}}{E_{partial}} \times 100\%. \quad (22)$$

Here, E serves as the placeholder for the error metrics MAE and RMSE. $E_{partial}$ are errors for the *partial* setting and E_{ours} represents the errors for GIMCC.

- **Oracle Improvement** ($\Delta_{improve}$). $\Delta_{improve}$ is used to measure the improvement of our method compared to the *Oracle* setting. Let E_{oracle} denote the error of the *Oracle* setting, the improvement $\Delta_{improve}$ can be represented as

$$\Delta_{improve} = \frac{E_{oracle} - E_{ours}}{E_{oracle}} \times 100\%. \quad (23)$$

For the improvement measurements Δ_{subset} and $\Delta_{improve}$, the larger the value, the better the performance.

4.1.5 Implementation Details. Our model and baselines were implemented using the PyTorch framework and evaluated on a Linux server equipped with a single 4090 GPU. In VSF, only a subset of variables, Ψ_S , is accessible during the inference phase. In this paper, both the historical time series length L and the prediction length Q are set as 12. For the selection of the variable subset, we set S to be some percent k of all variables N , $k = 15\%$ in this paper, aligning with the settings in [1, 10, 12]. We conduct 10 times training sessions, where each trained model is trained with 100 epochs. To ensure the Ψ_S 's broad coverage in Ψ_N , we randomly sample Ψ_S 100 times for each training session. Additionally, for the data segmentation, 70% of the samples are for training, 10% is for validation and 20% is for inference.

4.2 Overall Performance (RQ1)

Table 1 presents the overall performance of four time series forecasting models in different datasets. The results demonstrate that the integration of the proposed GIMCC method significantly enhances the final forecasting accuracy. In most cases, GIMCC leads to improvements of more than 20% in both MAE and RMSE, with some datasets exhibiting gains exceeding 50%. Notably, in the SOLAR dataset, GIMCC achieves a remarkable 60.27% improvement in MAE and 58.68% in RMSE, utilizing TGCN backbone. The last row of the table summarizes the average improvement across datasets, showing that GIMCC consistently enhances model performance, with most datasets benefiting from a 20% or greater improvement.

Overall, these results underscore the effectiveness of GIMCC in solving the Variable Subset Forecasting task.

4.3 Comparison with Oracle Setting (RQ2)

Table 2 provides a comprehensive comparison between our proposed model and the *Oracle* setting across different datasets. The results indicate that our model consistently outperforms the *Oracle* setting, achieving an average improvement of over 20%. These performance gains primarily result from our joint learning approach, which integrates the variable imputation model with the forecasting backbone. By leveraging the forecasting backbone to guide the imputation process, GIMCC mitigates potential errors or biases that traditional imputation methods might introduce, allowing the model to leverage both imputation and prediction synergistically for more accurate and reliable forecasts. To further validate the effectiveness of this joint learning module, we conduct additional experiments comparing models with and without joint learning, detailed in Section 4.7.

4.4 Comparison with Imputation Baselines on Variable Subset Forecasting Task (RQ3)

We compared the proposed method with ten imputation baselines and conducted a rigorous evaluation within a comprehensive imputation framework. From Figure 3 to Figure 4, GIMCC achieves optimal performance metrics on all datasets, where lower values indicate higher forecasting precision. The results clearly confirm the robustness and superiority of our imputation technique. The missingness of the entire variable can disrupt the intrinsic causal relationships between variables, thereby affecting the efficiency of imputation algorithms. Our method reconstructs these damaged causal associations, avoiding the uncertainty of direct variable recovery. It is capable of identifying and capturing the genuine relationships between variables, rather than mere statistical correlations, thus effectively promoting the predictive capability enhancement of the subsequent forecasting backbone.

4.5 Analysis of Global and Local Causality Consistency Constraints (RQ4)

To evaluate the effectiveness of the causality consistency module, we compare the performance replacing the learned causal matrix with other matrices. (1) **GIMCC-LM**: The learned causal graph is

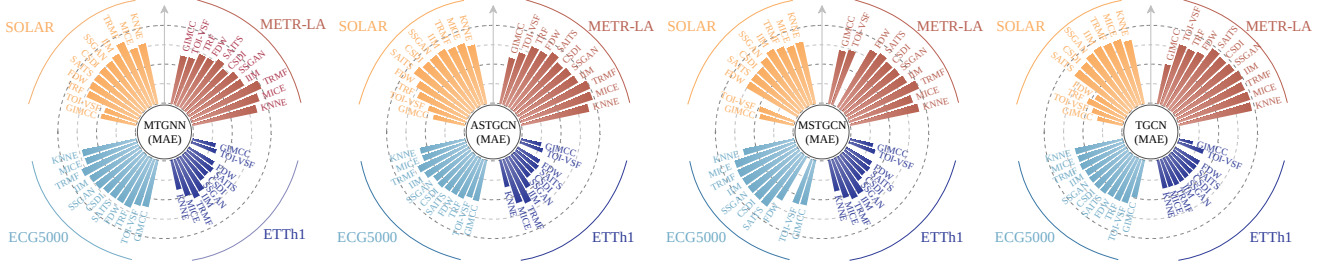


Figure 3: Performance of different imputation methods on VSF in MAE. The lower the results, the better the performance.

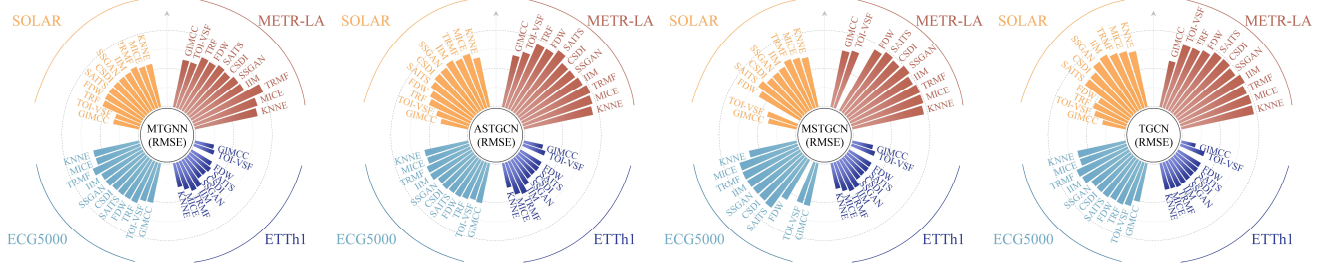


Figure 4: Performance of different imputation methods on VSF in RMSE. The lower the results, the better the performance.

Table 3: Performance Comparison on ECG5000 Dataset with Different Loss Weights. α , β and γ mean the proportion of imputation loss, global causality consistency loss and local subset-subgraph causality consistency loss, respectively. $\alpha + \beta + \gamma = 1$.

Forecasting Model	Weight Setting (α, β, γ)				
	1,0,0	0.2,0.5,0.3	0.3,0.4,0.3	0.5,0.25,0.25	0.8,0.1,0.1
MTGNN	3.39(0.42)/5.71(0.91)	3.15(0.51)/5.41(1.01)	3.14(0.51)/5.42(1.01)	3.14(0.50)/5.41(1.01)	3.15(0.52)/5.41(1.01)
ASTGCN	3.57(0.41)/5.82(0.93)	3.29(0.47)/5.58(0.98)	3.33(0.45)/5.58(0.95)	3.41(0.46)/5.64(0.95)	3.44(0.46)/5.67(0.95)
MSTGCN	3.14(0.51)/5.52(1.06)	3.13(0.51)/5.53(1.06)	3.12(0.51)/5.50(1.05)	3.14(0.51)/5.55(1.07)	3.15(0.52)/5.57(1.08)
TGCN	5.27(1.22)/8.51(2.33)	5.15(1.21)/8.34(2.27)	5.10(1.20)/8.28(2.26)	5.13(1.22)/8.36(2.30)	5.20(1.17)/8.34(2.21)

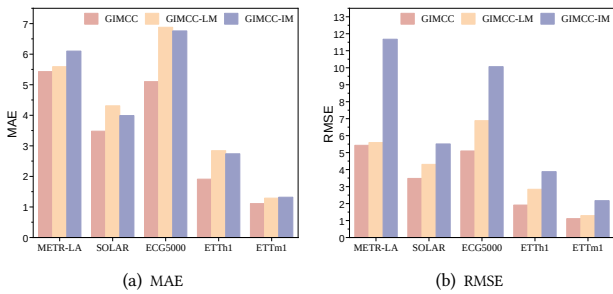


Figure 5: Ablation Study of causality consistency constraints in different datasets, utilizing the backbone TGCN. GIMCC-LM means the causal graph is replaced with a learnable adjacency matrix. GIMCC-IM means the causal graph is replaced with an identity matrix.

replaced with a learnable adjacency matrix. (2) **GIMCC-IM**: The learned causal graph is replaced with an identity matrix.

Figure 5 demonstrates that replacing the learned causal graph leads to a significant decline in overall performance, underscoring

the importance of maintaining both global and local causality consistency. This finding suggests that the adjacency matrix alone fails to accurately capture the underlying relationships within the data. Furthermore, as shown in Table 3, removing the loss terms associated with these two levels of causality consistency also leads to a substantial drop in performance. Therefore, ensuring causality consistency allows the imputed data to align with the true underlying causal mechanisms, ultimately improving the prediction precision. Additional experimental results can be found in Appendix B.

4.6 Impact on Weights Variation of Loss on Forecasting Performance (RQ5)

In this section, we systematically evaluate the impact of different weight settings on model performance. Specifically, we set $\alpha + \beta + \gamma = 1$ and vary the combinations of α , β and γ , which represents the parameters in Equation 21. As shown in Table 3, the results indicate that all three loss terms play a crucial role in the model's performance. And when the weight setting vary, we can observe a clear trend in the changes to the final forecasting performance. Notably, when the three loss weights are approximately

Table 4: Ablation Study of Joint Learning. ‘w/o JL’ means the imputation model undergoes solely pre-training prior to the training of the forecasting model.

Forecasting Model		METR-LA		SOLAR		ECG5000		ETTh1		ETTm1	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
ASTGCN	w/o JL	3.98(0.27)	7.69(0.45)	4.28(0.34)	5.78(0.60)	3.10(0.49)	5.32(0.98)	1.66(0.97)	2.76(1.78)	1.02(0.59)	1.71(1.07)
	joint learning	3.74(0.25)	7.31(0.43)	2.72(0.22)	3.96(0.43)	3.29(0.47)	5.58(0.98)	1.61(0.93)	2.69(1.71)	0.98(0.57)	1.68(1.06)
MSTGCN	w/o JL	3.84(0.26)	7.61(0.45)	3.05(0.24)	4.35(0.50)	3.38(0.56)	5.72(1.10)	1.63(0.93)	2.70(1.69)	1.00(0.56)	1.68(1.03)
	joint learning	3.72(0.26)	7.35(0.45)	2.38(0.22)	3.69(0.43)	3.12(0.51)	5.50(1.05)	1.60(0.94)	2.66(1.70)	0.97(0.56)	1.67(1.05)
MTGNN	w/o JL	3.97(0.27)	7.90(0.48)	3.14(0.26)	4.43(0.45)	3.42(0.47)	5.75(0.98)	1.68(0.96)	2.82(1.79)	0.97(0.54)	1.64(1.00)
	joint learning	3.25(0.23)	6.55(0.43)	2.38(0.22)	3.58(0.41)	3.14(0.50)	5.41(1.01)	1.57(0.89)	2.64(1.64)	0.96(0.55)	1.65(1.03)
TGCN	w/o JL	8.49(0.83)	14.31(1.14)	5.57(0.71)	7.77(1.29)	6.23(1.47)	9.88(2.48)	2.99(1.54)	4.57(2.63)	1.29(0.77)	2.08(1.37)
	joint learning	5.43(0.40)	10.03(0.58)	3.48(0.36)	5.02(0.60)	5.10(1.20)	8.28(2.26)	1.91(0.96)	3.06(1.84)	1.11(0.59)	1.82(1.12)

equal, the model achieves optimal performance in both forecasting tasks and causal relationship construction, striking an effective balance. This balance ensures the coordination between forecasting tasks and causal structure capture while maximizing the advantages of each component, achieving robust model performance. However, the optimal weight configuration may exhibit slight variations depending on the specific forecasting backbone and dataset used. For additional supplementary experiments, please refer to Appendix C.

4.7 Analysis of Joint Learning (RQ6)

We compare the performance of the models with and without joint learning to assess the impact of joint learning on multi-task optimization in Table 4.

- **Without Joint Learning (w/o JL):** In this setup, the imputation model is pre-trained independently before the training of the forecasting model begins. This approach treats the tasks sequentially, without leveraging the synergies between them.

These results highlight the effectiveness of joint learning in optimizing multiple tasks simultaneously. For instance, on the METR-LA dataset, the average performance metric of the backbone model with joint learning was significantly higher than that of the model trained without joint learning. This demonstrates that joint learning can effectively enhance the overall performance of the model by optimizing multiple tasks concurrently.

5 Related Work

5.1 Time Series Imputation

The presence of missing data can significantly degrade the performance of downstream tasks, such as prediction and classification. Therefore, when designing models, it is crucial to consider the impact of missing data on overall performance. Missing data imputation methods can be categorized into several types, including simple value imputation [9], neighbor-based imputation [1, 2], machine learning-based imputation [10, 33, 37], or deep learning-based imputation [3, 12, 31]. Additionally, adversarial networks have been adapted for the imputation tasks [19, 36]. While these approaches have proven effective in data imputation tasks, they ignored the unavailable correlations between variables in VSF and distribution shifts over time. Our approach uses two causality consistency terms to guide the imputation process.

5.2 Causal Discovery

Causal discovery is to recognize the causal relations between variables [5]. The learned causal graph could be beneficial for the data imputation. Existing work can be divided into four categories: constraint-based methods, score-based methods, function causal model (FCM)-based methods [13, 21, 25] and Granger causality-based methods. Constraint-based approaches depend on statistical tests of conditional independence and construct the DAG between variables, such as oCSE [29] and PCMCI [26]. Score-based methods [4, 16, 24] do not rely on a predefined causal structure, which can flexibly adapt to different data structures and distributions. Granger causality [6] is a popular tool for analyzing time series data [5]. Recent advances of Granger causality are based on neural networks, which can represent nonlinear and complex interactions between variables, like self-explaining networks [18], attention mechanisms [20] and variational autoencoders [17].

6 Conclusion Remarks

In this paper, we addressed the Variable Subset Forecasting (VSF) problem by introducing Generative Imputation with Multi-level Causal Consistency (GIMCC). By leveraging causality as a guiding principle, GIMCC effectively overcomes the limitations of traditional correlation-based imputation methods, such as unavailable correlations and distribution shifts over time. Our generative imputation method ensures that imputed data satisfies both global and local causal consistency constraints, preserving the underlying causal relationships. Furthermore, the joint learning strategy, which uses the forecasting backbone to guide the imputation model, enhances predictive performance. Extensive experiments on real-world datasets validate the effectiveness of our approach, demonstrating significant improvements in both imputation and forecasting accuracy for the VSF task. Our findings highlight the potential of causality in advancing the state of the art in time series forecasting under the constraints of substantial missing data.

Acknowledgments

This research is funded by the Science and Technology Development Fund (FDCT), Macau SAR (file no. 0123/2023/RIA2, 001/2024/SKL).

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A Experimental Details

A.1 Dataset Details

We conduct our experiments on the following five real-world datasets:

- **METR-LA** dataset covers the average traffic speed of the highways in Los Angeles County, collected from 207 loop detectors from Mar 2012 to Jun 2012.
- **SOLAR** dataset records the solar power output collected from 137 plants in Alabama State in 2007.
- **ECG5000** dataset, sourced from the UCR Time-Series Classification Archive, comprises 140 electrocardiograms (ECG) each with a length of 5000. Similar to the Traffic dataset, we increase the scale of all variables by a factor of 10.
- **ETTh1**, **ETTm1** datasets contain the oil temperature and six types of power load features from electricity transformers, collected from July 2016 to July 2018. The **ETTh1** dataset is recorded hourly, while the **ETTm1** dataset is recorded every 15 minutes.

A.2 Backbone Details

In this paper, we choose the following widely used forecasting models as the backbone models to represent the diverse range of recent techniques.

- **MTGNN** [34]: The model utilizes a graph learning module to extract one-way relationships between variables, followed by joint learning of spatio-temporal relationships for spatio-temporal modelling.
- **ASTGCN** [8]: This approach incorporates adaptive weighting through the use of advanced spatiotemporal attention mechanisms, which are designed to dynamically discern and learn the complexities of spatiotemporal relationships, thereby enhancing the accuracy and efficacy of predictive models.
- **MSTGCN** [8]: The approach utilizes a multiscale graph convolutional network to capture spatial features and employs temporal convolution along the time dimension to capture temporal dependencies.
- **TGCN** [39]: This method introduces temporal feature learning to capture the evolution of time features and employs graph convolutional networks to capture the spatio-temporal relationships between nodes.

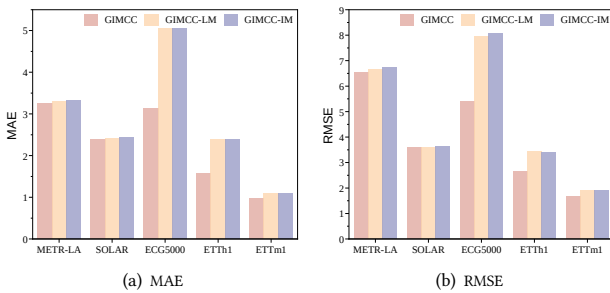


Figure 6: Ablation study of causality consistency constraints in different datasets, utilizing the backbone MTGNN. GIMCC-LM means the causal graph is replaced with a learnable adjacency matrix. GIMCC-IM means the causal graph is replaced with an identity matrix.

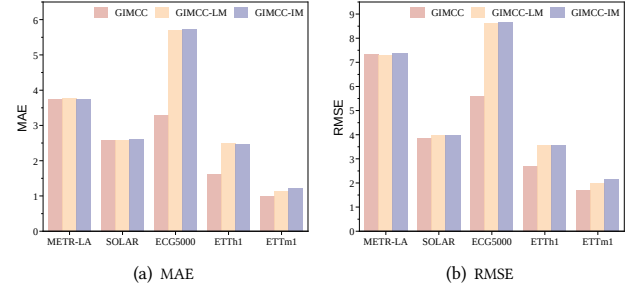


Figure 7: Ablation Study of causality consistency constraints in different datasets, utilizing the backbone ASTGCN. GIMCC-LM means the causal graph is replaced with a learnable adjacency matrix. GIMCC-IM means the causal graph is replaced with an identity matrix.

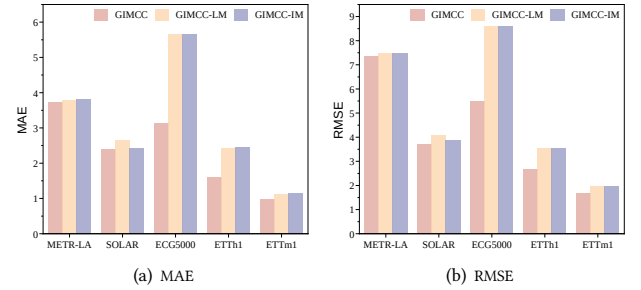


Figure 8: Ablation Study of Causality Consistency constraints in different datasets, utilizing the backbone MSTGCN. GIMCC-LM means the causal graph is replaced with a learnable adjacency matrix. GIMCC-IM means the causal graph is replaced with an identity matrix.

B Supplementary Results of Ablation Study for Causality Consistency (RQ4)

We have supplemented the experimental results on the backbone MTGNN, ASTGCN and MSTGCN. As shown in Figure 6 to Figure 8, GIMCC achieves the best forecasting performance, which indicates the importance of causality consistency terms.

C Supplementary Results of Weights Variation of Loss on Performance (RQ5)

In this section, we have supplemented the experimental results of the weight variation of loss on forecasting performance on the METR-LA, SOLAR, ETTh1 and ETTm1 datasets.

D More Forecasting Backbone Results

We have included additional forecasting backbones (TSMixer, TimesNet and Informer) to provide a more comprehensive evaluation. As shown in Table 6, irrespective of the specific forecasting models applied, GIMCC leads to a consistent enhancement in forecasting accuracy on dataset ECG5000.

E Model Complexity Analysis

Here, we provide a detailed analysis of the model complexity based on the provided information. The causal discovery phase employs

Table 5: Performance Comparison on SOLAR, METR-LA, ETTh1, and ETTm1 Datasets with Different Loss Weights.

Forecasting Model	SOLAR Weight Setting (α, β, γ)				
	1,0,0	0.2,0.5,0.3	0.3,0.4,0.3	0.5,0.25,0.25	0.8,0.1,0.1
MTGNN	2.38(0.21)/3.65(0.42)	2.38(0.22)/3.58(0.41)	3.31(0.48)/5.64(0.97)	3.41(0.28)/4.70(0.46)	2.67(0.26)/4.08(0.50)
ASTGCN	3.26(0.22)/6.60(0.41)	3.26(0.23)/6.58(0.42)	3.27(0.23)/6.58(0.43)	3.25(0.23)/6.55(0.43)	3.25(0.23)/6.56(0.43)
MSTGCN	1.60(0.89)/2.67(1.65)	1.57(0.89)/2.64(1.65)	1.57(0.89)/2.64(1.64)	1.59(0.89)/2.67(1.65)	1.57(0.90)/2.64(1.66)
TGCN	0.97(0.56)/1.66(1.05)	0.96(0.56)/1.64(1.03)	0.96(0.55)/1.65(1.03)	0.96(0.56)/1.64(1.04)	0.96(0.55)/1.64(1.03)
Forecasting Model	METR-LA Weight Setting (α, β, γ)				
	1,0,0	0.2,0.5,0.3	0.3,0.4,0.3	0.5,0.25,0.25	0.8,0.1,0.1
MTGNN	2.59(0.23)/3.89(0.46)	2.68(0.22)/3.93(0.42)	3.86(0.25)/5.00(0.43)	3.87(0.26)/5.10(0.44)	2.57(0.23)/3.86(0.45)
ASTGCN	3.73(0.24)/7.31(0.42)	3.74(0.25)/7.32(0.43)	3.74(0.25)/7.31(0.43)	3.75(0.25)/7.32(0.43)	3.74(0.25)/7.32(0.43)
MSTGCN	1.61(0.93)/2.68(1.71)	1.61(0.94)/2.69(1.72)	1.62(0.93)/2.69(1.71)	1.61(0.93)/2.69(1.71)	1.62(0.94)/2.70(1.72)
TGCN	0.99(0.57)/1.68(1.06)	0.98(0.57)/1.68(1.06)	0.99(0.57)/1.68(1.07)	0.98(0.57)/1.68(1.07)	0.99(0.57)/1.68(1.07)
Forecasting Model	ETTh1 Weight Setting (α, β, γ)				
	1,0,0	0.2,0.5,0.3	0.3,0.4,0.3	0.5,0.25,0.25	0.8,0.1,0.1
MTGNN	2.38(0.20)/3.71(0.43)	2.64(0.24)/4.05(0.46)	2.39(0.22)/3.72(0.44)	2.38(0.22)/3.69(0.43)	2.80(0.31)/4.18(0.58)
ASTGCN	3.72(0.26)/7.35(0.45)	3.74(0.26)/7.36(0.45)	3.72(0.26)/7.35(0.45)	3.72(0.26)/7.34(0.45)	3.72(0.26)/7.36(0.45)
MSTGCN	1.60(0.93)/2.66(1.70)	1.63(0.95)/2.70(1.71)	1.60(0.94)/2.66(1.70)	1.60(0.94)/2.66(1.70)	1.60(0.94)/2.66(1.70)
TGCN	0.98(0.57)/1.67(1.06)	0.97(0.56)/1.67(1.05)	0.98(0.57)/1.67(1.06)	0.97(0.56)/1.67(1.05)	0.98(0.57)/1.67(1.06)
Forecasting Model	ETTh1 Weight Setting (α, β, γ)				
	1,0,0	0.2,0.5,0.3	0.3,0.4,0.3	0.5,0.25,0.25	0.8,0.1,0.1
MTGNN	5.17(0.29)/6.69(0.55)	3.48(0.36)/5.02(0.60)	5.23(0.37)/6.72(0.60)	7.24(0.41)/8.80(0.62)	8.57(0.40)/10.15(0.60)
ASTGCN	6.06(0.42)/11.39(0.61)	5.68(0.37)/10.59(0.56)	5.92(0.50)/10.11(0.65)	7.70(0.58)/13.49(0.78)	5.43(0.40)/10.03(0.58)
MSTGCN	1.88(0.96)/3.02(1.84)	1.94(1.04)/3.11(1.94)	1.93(1.05)/3.11(1.96)	1.91(0.96)/3.06(1.84)	1.94(1.06)/3.13(1.97)
TGCN	1.15(0.57)/1.84(1.09)	1.12(0.58)/1.83(1.11)	1.11(0.59)/1.82(1.12)	1.13(0.59)/1.84(1.11)	1.16(0.57)/1.85(1.09)

Table 6: More forecasting backbone results on ECG5000 dataset. The results are reported as an average and standard deviations (values in parenthesis) of 10 runs. Each run corresponds to one trained model with 100 epochs, and each epoch involves a randomly sampled variable subset. Δ_{subset} represents the improvement of our model over the *Partial* setting. The larger the value, the better the performance for the Δ_{subset} .

Backbone	TSMixer		TimesNet		Informer	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
Partial	3.14(0.53)	5.36(1.02)	3.07(0.48)	5.29(1.00)	7.24(1.37)	10.88(2.26)
Ours	3.06(0.47)	5.27(0.99)	3.03(0.52)	5.27(1.04)	3.49(0.65)	5.94(1.15)

separate multi-layer perceptrons (MLPs) for each variable, where the weights of these MLPs are utilized to infer causal relationships. The training time complexity for a single MLP is defined as $O(E \cdot D \cdot L \cdot h^2)$, with E , D , L , and h representing the number of training epochs, samples, layers, and hidden nodes per layer, respectively. For a system with N variables, each requiring an independent MLP, the total training complexity scales to $O(N \cdot E \cdot D \cdot L \cdot h^2)$, reflecting a linear dependence on the number of variables.

During inference, subset data comprising S variables and a time-domain sequence of length T are transformed into the Fourier domain and convolved with the frequency representation of the complete graph. The subset Fourier transform step incurs a complexity of $O(S \cdot T \log T)$, while the subsequent graph Fourier transform and convolution, which embeds the subset into an N -dimensional space via zero-padding, requires $O(N^2)$ operations. The total per-sample complexity for this phase is $O(S \cdot T \log T + N^2)$, with the N^2 term dominating when the subset size S is significantly smaller than the total number of variables N .